

Airline Delays: Operational Weaknesses and Financial Patterns in U.S. Airlines

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Introduction

Airline delays are one of the most persistent operational challenges in the aviation industry, affecting millions of passengers annually and imposing substantial costs on airlines, airports, and the broader economy. This project was inspired by Seira Ramchandani's career interest in aviation operations and Graham Blechschmidt's interest in the application of Economics in non-traditional settings or industries. This project ultimately worked as an effort to explore these curiosities while simultaneously developing technical skills in Python and its data manipulation libraries.

Background

The main dataset used for the project was the Airline_Delay_Cause dataset, obtained from Kaggle and originally collected via the Department of Transportation. The supplementary data set was made using the yfinance library. Specifically, it pulled the stock values (from the NASDAQ) for the first of every month from 2018 to 2023. If the first of the month fell on a date on which the stock market was closed, it would pull the most recent available value. The chosen companies were those from the initial dataset carrier list that were public and available as tradable companies.

Research Questions

Two central research questions guided the analysis:

1. **How does the stock value compare to the delay time of airlines?**
2. **What could be done operationally to decrease the amount of delays for the 5 worst airlines based on the observed data patterns of the best 5?**

The first research question looks into the factors that determine the economic prosperity of the airlines. It acknowledges that these are companies with a profit-driven mindset and works to determine the connection between actual performance (delay time) and the value. The second research question focuses on the actions and reasons behind high-performing airlines (those with fewer delays), it focuses on how there can be improvement for delay times, and looks to highlight the best operational practices. Although operational disruptions are usually studied in isolation from stock market valuation, we view them as potential interconnected components of airline performance.

Methods

The variables we focused on were: carrier_name, carrier_delay, delay_rate (derived from delay_15mins_count / arriving_flights_counts), avg_delay_minutes (derived from total_delay_arriving_flights_counts), weather_count, carrier_count, nas_count, security_count, late_aircraft_count, and stock_valuation (made using the available data from yfinance). For each airline and airport, the count variables produced a sum of the number of occurrences in each operational delay cause, and how many minutes of delay came from different operational causes. Delay rate is the proportion of flights delayed, and was computed to normalize across different-sized airlines and observations. Similarly, Average delay minutes captures the typical delay burden per flight. Lastly, the stock value variable was in USD.

The Airline_Delay_Cause dataset contained data from 2013 to 2023 with over 170,000 rows. We filtered it to only include the 5 most recent years (2018-), and cleaned the data by renaming columns for ease of use and creating a date column for ease of plotting.

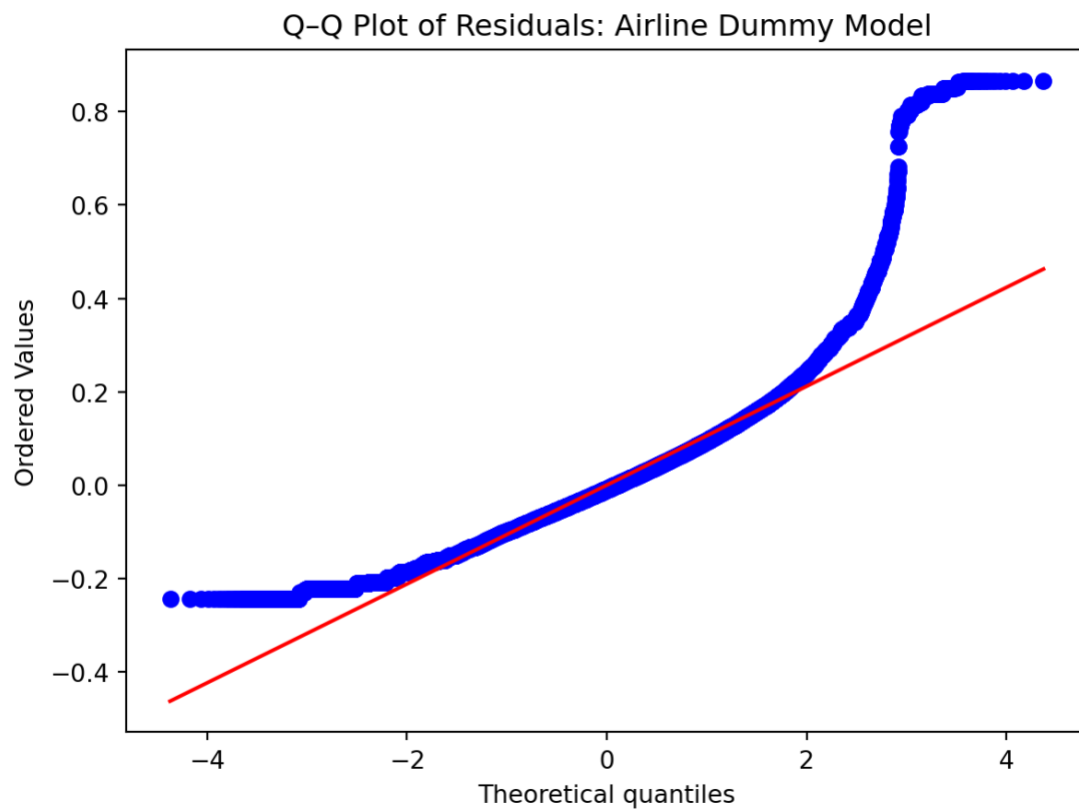
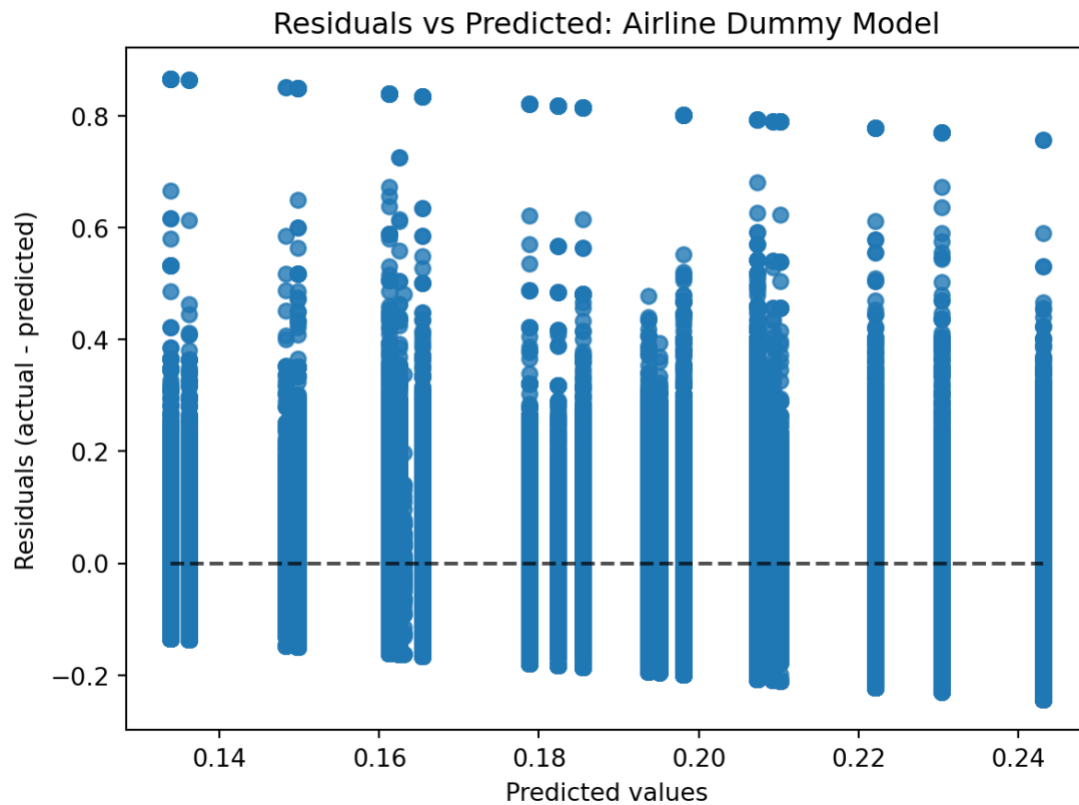
There was minimal missingness found in the dataset; only 0.15% of 112690 rows had missing data. A heatmap of missingness correlation values showed that if an observation was missing for one variable, it was missing for all the other variables. Additionally, checks for missingness based on variables such as airline and airport gave no results. We concluded that our data was missing completely at random (MCAR) and could be dropped.

To assess whether airline identity or basic operational variables predict delay outcomes, 2 models were developed: (1) a simple linear regression using airline dummy variables with a LASSO regression for feature selection, and (2) a multiple linear regression model using month and flight volume.

Model 1: Does the Airline Predict Delay Rate?

R² score (simple linear regression): 0.06755123845520838

	Airline	Coefficient
10	JetBlue Airways	0.080627
8	Hawaiian Airlines Inc.	0.067866
7	Frontier Airlines Inc.	0.059520
5	ExpressJet Airlines Inc.	0.047611
1	American Airlines Inc.	0.046722
0	Allegiant Air	0.044795
17	United Air Lines Inc.	0.035505
15	Southwest Airlines Co.	0.032555
16	Spirit Air Lines	0.031160
11	Mesa Airlines Inc.	0.022940
6	ExpressJet Airlines LLC	0.019816
12	PSA Airlines Inc.	0.016176
4	Envoy Air	0.002832
18	Virgin America	0.000491
14	SkyWest Airlines Inc.	-0.001356
13	Republic Airline	-0.012814
9	Horizon Air	-0.014315
2	Delta Air Lines Inc.	-0.026427
3	Endeavor Air Inc.	-0.028762



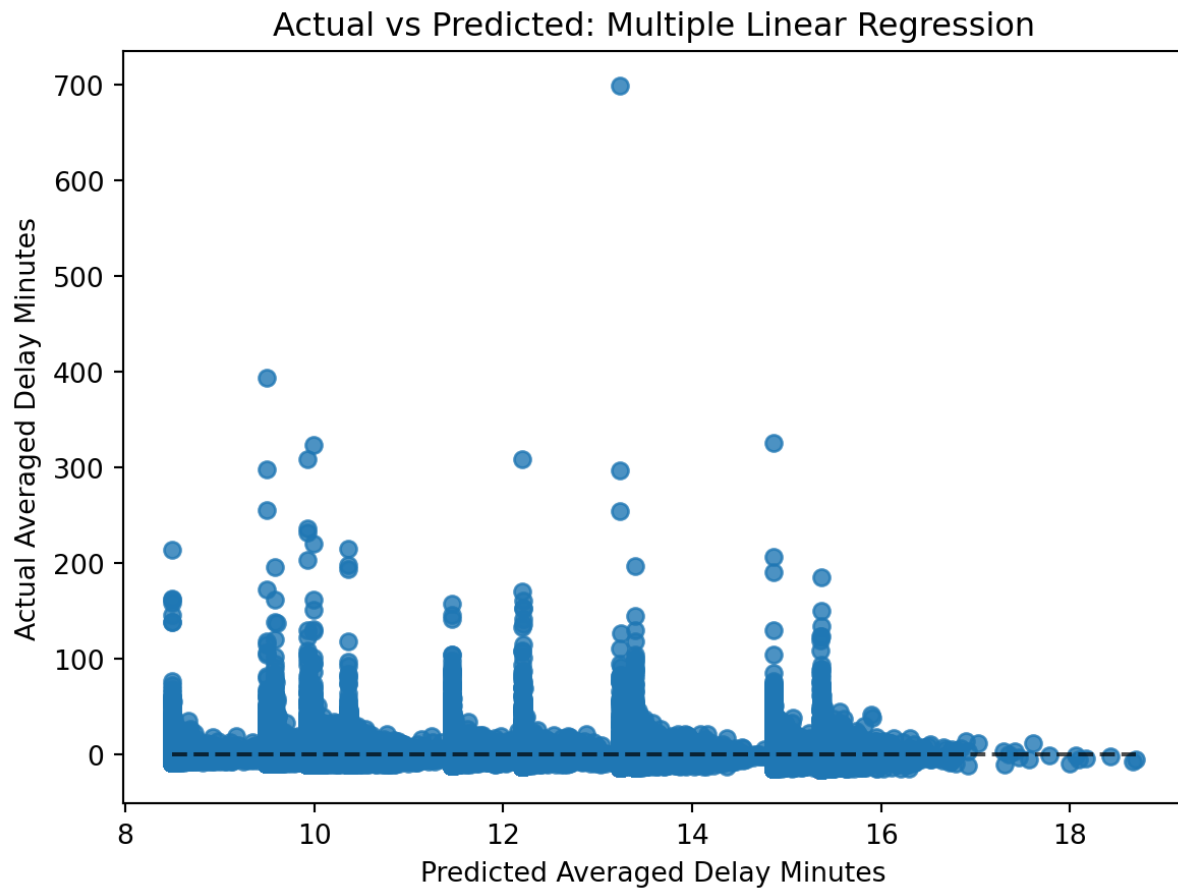
R² Score for LASSO regression model: 0.0

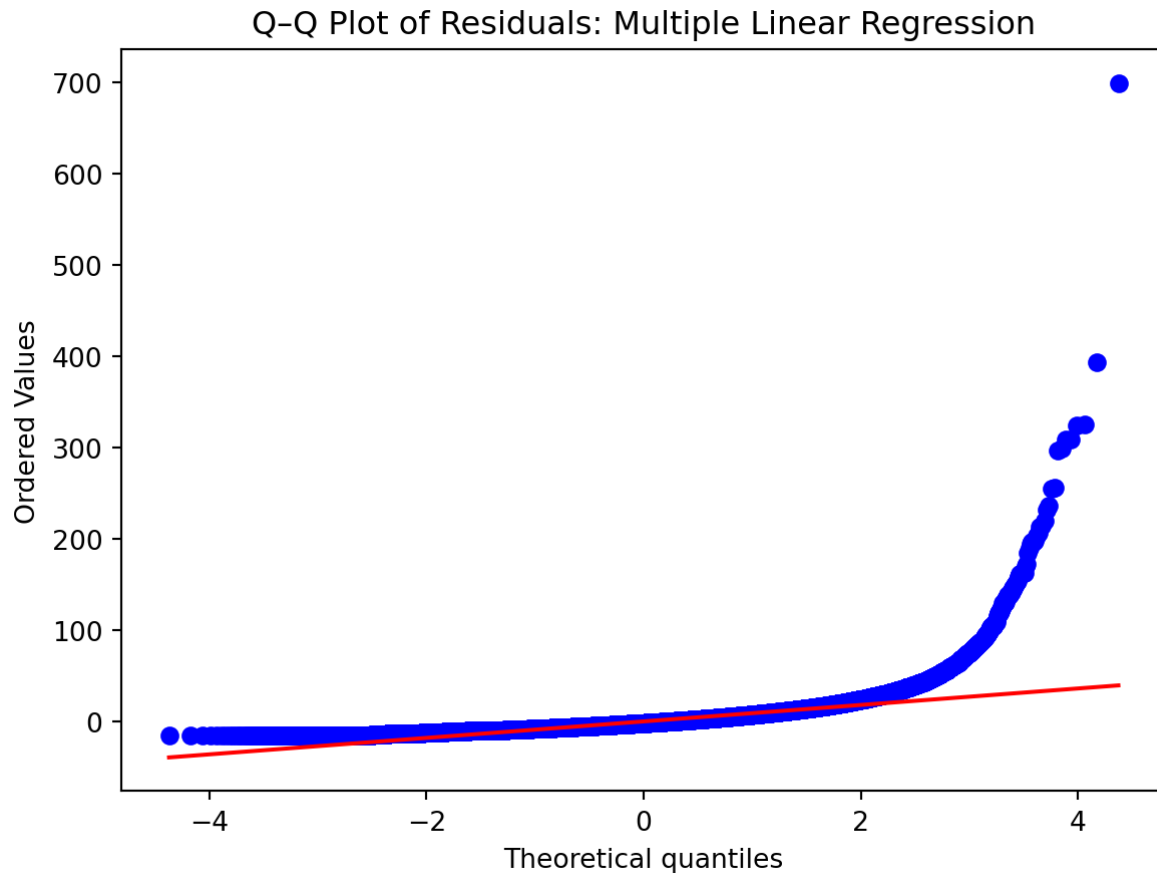
Coefficients varied across airlines but were small and noisy, suggesting no strong or consistent pattern. Additionally, the diagnostic plots showed non-normal skewed residuals and wide scatter around the prediction line, proving poor overall model fit. Residuals vs. predicted-value plots revealed heteroscedasticity and violated the constant-variance assumptions.

The model's R^2 was extremely low (0.0675), and a LASSO model reduced all coefficients to zero, yielding an R^2 of 0.0. This signaled that **individual airline identity is insufficient for predicting delay rate.**

Model 2: Do Month and Flight Volume Predict Delay Minutes?

R^2 score (multi linear regression): 0.03969501268101028





The actual vs predicted plot shows that the multiple linear regression model predicts average delay minutes in a relatively narrow range (approximately 9–18 minutes), while the actual values span a much wider range (0–700 minutes). Most points fall above the reference line, suggesting that the model underpredicts large delays but captures the general operational trends. Q–Q plots again confirmed non-normal residuals, reinforcing that the assumption of linearity for this model is not met.

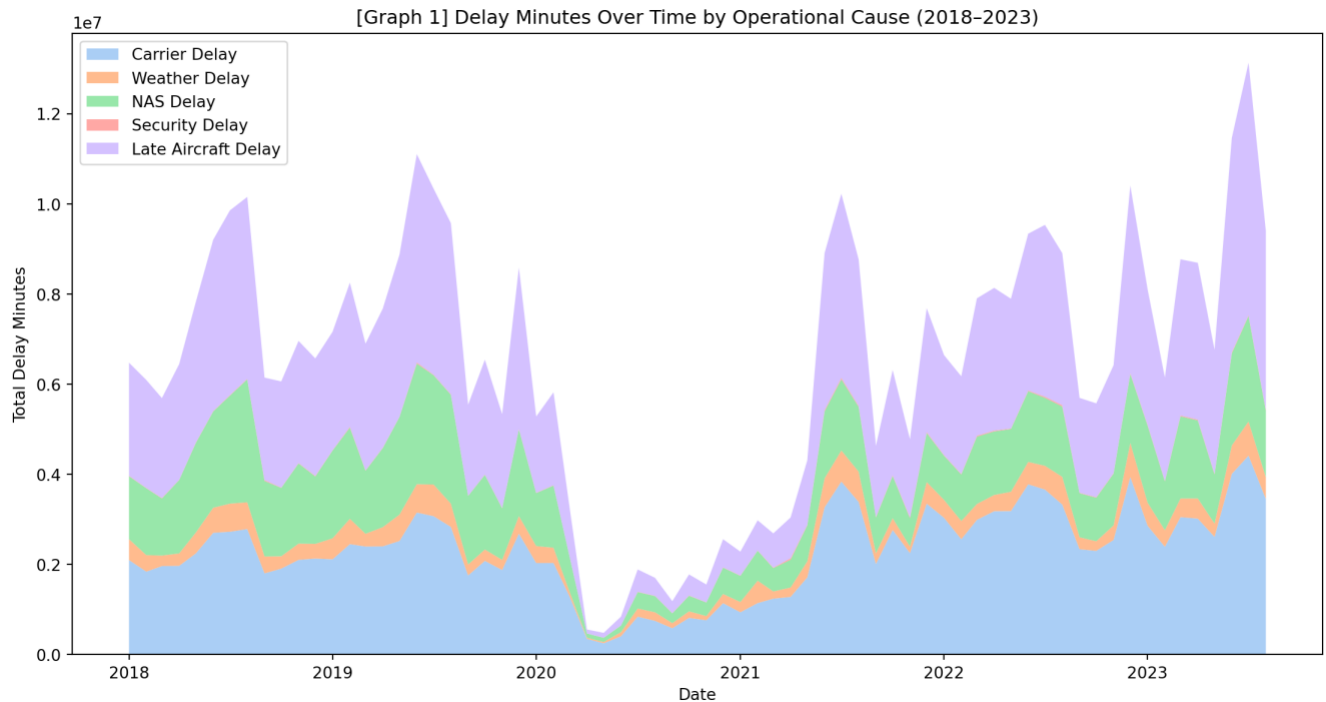
* Out of interest, the extreme outlier was a flight to Long Beach, CA, in December of 2021 with Envoy Air, and the delay was mostly due to the weather.

The model produced an extremely low R^2 (~ 0.04), indicating that **month and flight volume do not meaningfully predict delay outcomes on their own.**

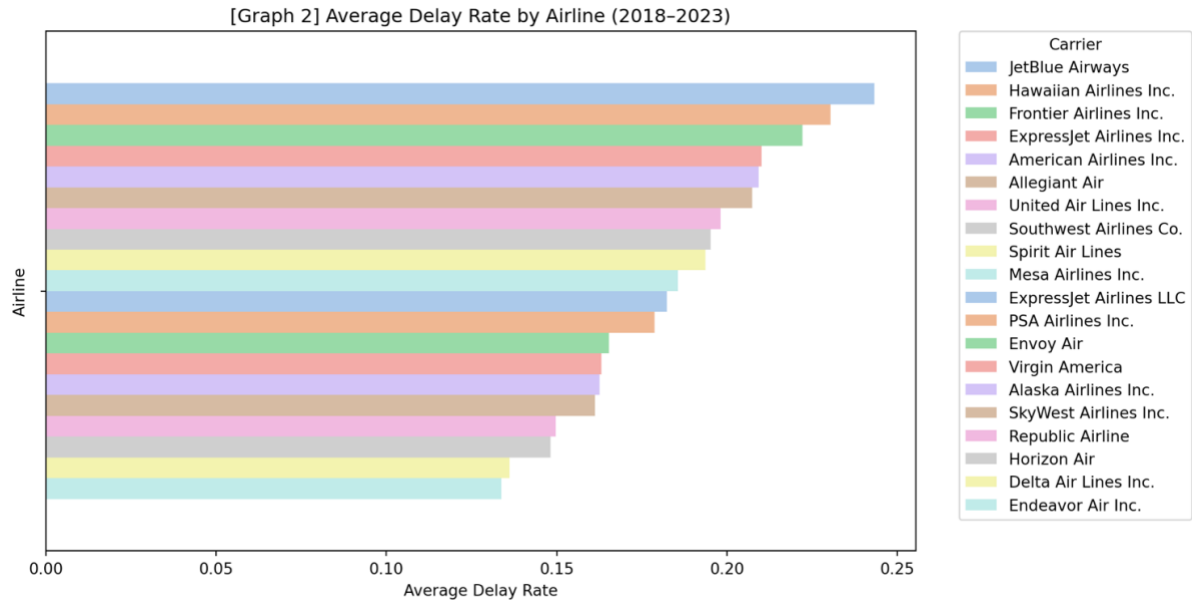
Overall, *linear regression was not the right model to fit the data.* However, the models suggest that **delays are driven by complex interactions, not simple predictors, and performance differences arise from operational behavior.**

Analysis & Results

To assess big-picture patterns in delay behavior, Graph 1 looks at total delay minutes over time by operational cause. The chart reveals that **late aircraft delays** dominate across nearly all months, with **carrier delays** being the second-largest contributor.

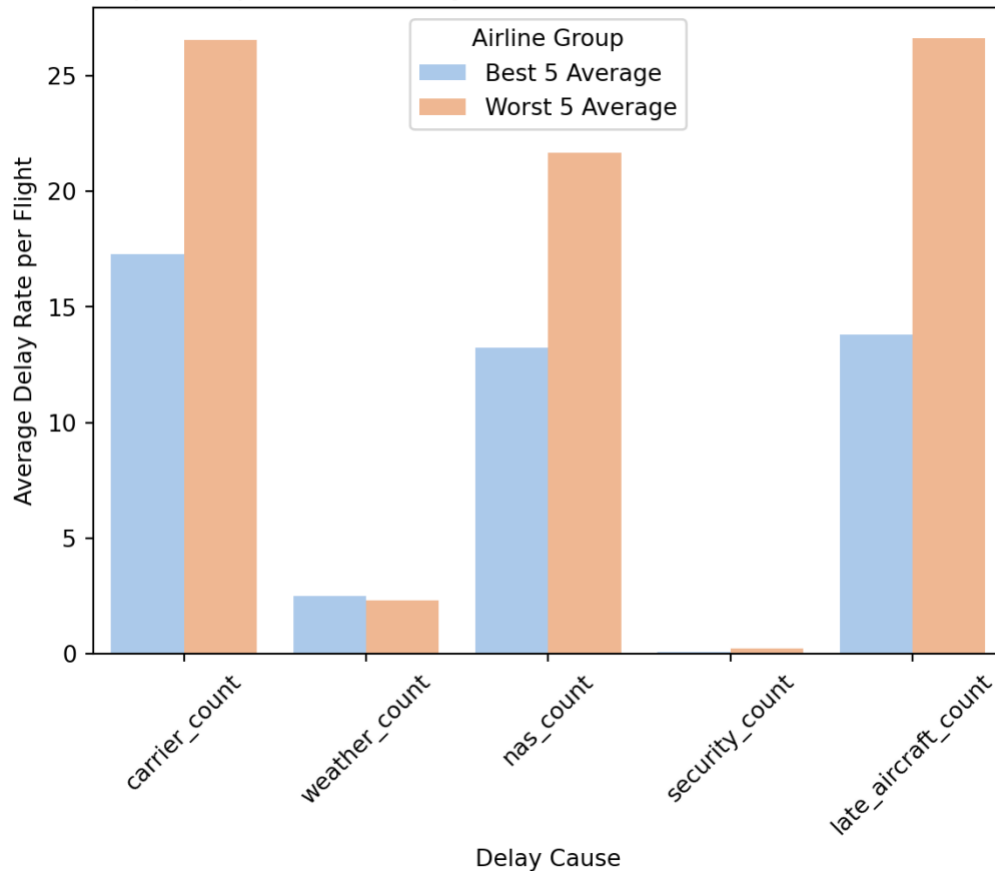


Next, by ranking airlines according to delay rate, we identified the best five and worst five performing airlines (Graph 2). Although the spread in delay rate is modest (roughly 14–24%), the operational differences between the top and bottom groups are clear in Graph 3. Worst airlines exhibit dramatically higher delay minutes across nearly all operational categories, especially carrier, NAS, and late aircraft delays. These trends suggest that **delays stem primarily from operational design and resilience, not random environmental events.**

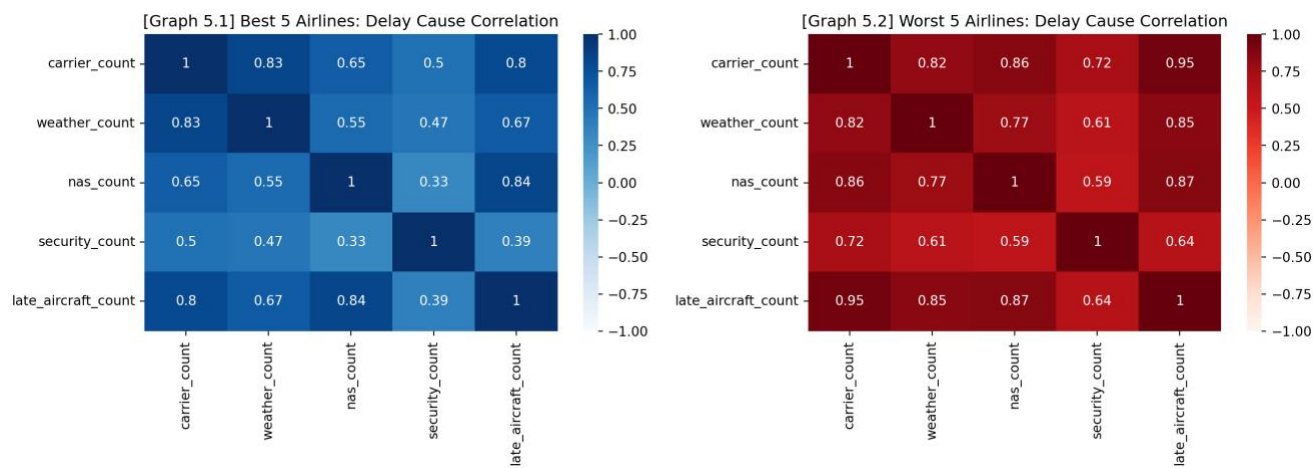


This graph indicates that the best 5 airlines with the shortest delays on average are: ['JetBlue Airways', 'Hawaiian Airlines Inc.', 'Frontier Airlines Inc.', 'ExpressJet Airlines Inc.', 'American Airlines Inc.'], and the worst 5 airlines are: ['SkyWest Airlines Inc.', 'Republic Airline', 'Horizon Air', 'Delta Air Lines Inc.', 'Endeavor Air Inc.']. .

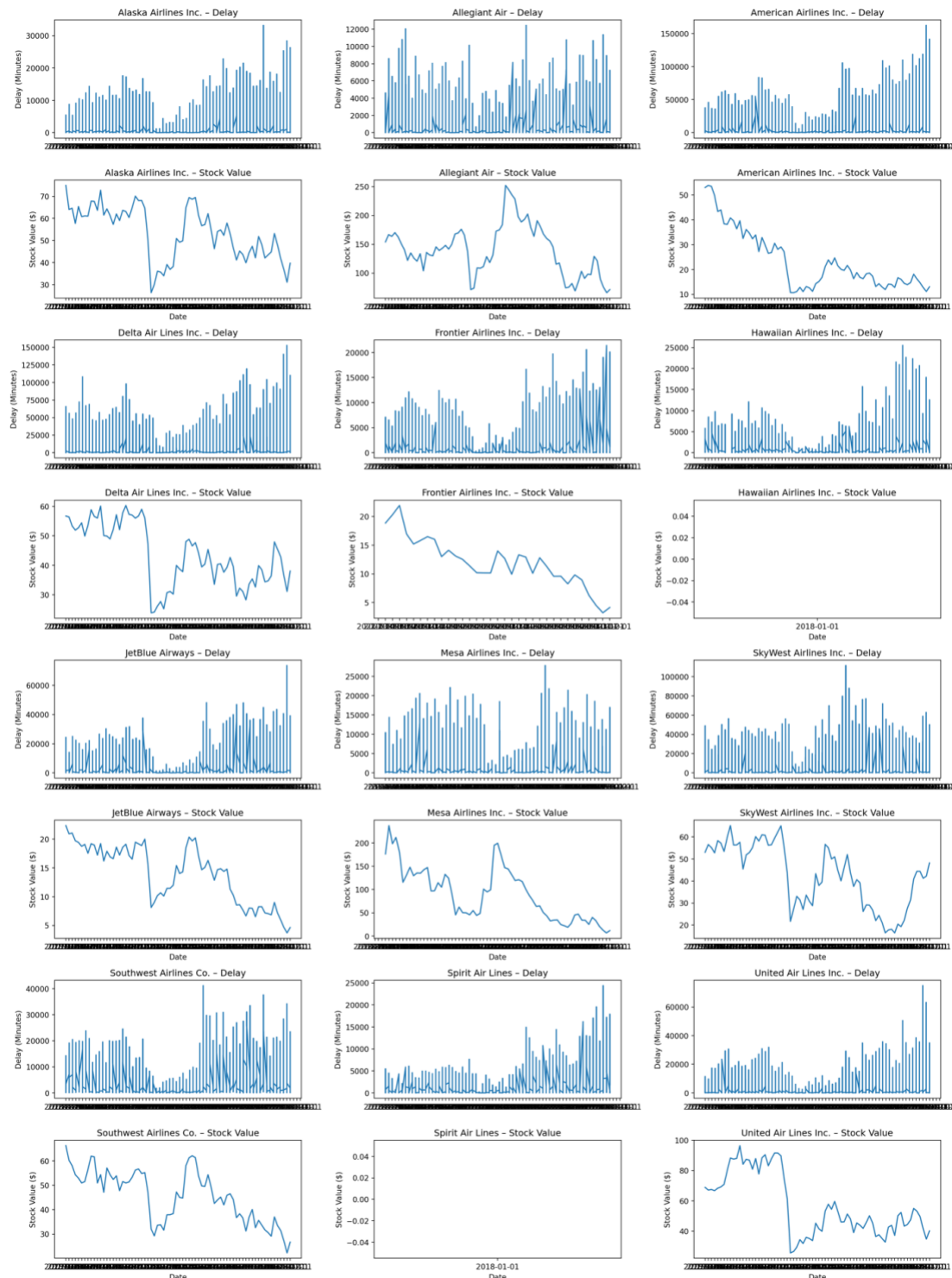
[Graph 3] Operational Delay Cause Rates: Best 5 vs Worst 5 Airlines



Heatmaps of operational delay correlations (Graph 4) indicate that for the best airlines, operational delay causes are moderately correlated. In contrast, the worst airlines show correlations approaching 0.9 or higher among multiple causes, suggesting that operational failures have a ripple effect within their systems.



Next, the influence of operational delays on airline stock values was explored. The following graph displays the stock value of the public airlines over the period of January 1st, 2018, to January 1st, 2023. The overall trend expresses a higher value in the initial years, a significant dip during 2020, and then a varied rebound of the different stock values. Overall, **the values tended to be lower than the initial start values in 2018.**



carrier_name	
Frontier Airlines Inc.	-0.125963
JetBlue Airways	-0.045700
Delta Air Lines Inc.	-0.023114
United Air Lines Inc.	-0.008794
SkyWest Airlines Inc.	-0.006448
American Airlines Inc.	-0.000541
Allegiant Air	0.025818
Alaska Airlines Inc.	0.042644
Mesa Airlines Inc.	0.070571
Southwest Airlines Co.	0.080409
Hawaiian Airlines Inc.	NaN
Spirit Air Lines	NaN

dtype: float64

The faceted graph displays the delay (minutes) attributed to a carrier alongside the stock value of that carrier. Both graphs are on a timeline spanning from January 1st, 2018, to January 1st, 2023. Overall, the graphs tend to show a visual **inverse relationship between the delay time and the stock value** during that same period. American Airlines is an excellent example of this, but some stray away from this pattern, such as Mesa Airlines, which is more variable overall.

Accompanying these graphs is a set of correlation values between each of the carrier delay and the stock valuation. The largest negative correlation was -.126 for Frontier Airlines. The highest positive correlation was .080 for Southwest Airlines. All the other airlines fell between these two values.

Conclusion

The worst-performing airlines have a significantly higher delay time in late aircraft delays and carrier delays, and their operational issues tend to cascade across the network. The machine learning models confirmed that airline identity alone is not enough to predict delays, reinforcing that it's **operational behavior that drives performance**. Despite the assumed visual relationship between stock value and carrier delay, *there was no significant correlation*. This articulates that the **primary driver of the value is less focused on physical performance (delay times)** and could be attributed to generalized economic trends or priorities that the timeline fits with.

Caveats

- Types of weather causing delays and airport congestion data were not available
- National Airspace System (NAS) Operations could have biases for some airlines and do not directly link to airline-specific operational capabilities
- Stock data is limited to publicly traded airlines
- Stock data does not display dividend or subsidiary relationships
- Stock data was taken on the 1st of the month, and may miss day-to-day trends